

Alessandro Petitti

ROBOTICS AND CONTROL ENGINEER

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Education

ETH - Eidgenössische Technische Hochschule

Zürich, Switzerland

MSC IN ROBOTICS SYSTEMS AND CONTROLS

Sep 2024 - present

- GPA: 5.6/6
- Relevant coursework: Physically based Simulation in computer graphics, Machine Perception, Vision Algorithms for Mobile Robotics, Robot Dynamics, Nonlinear Dynamics and Control, Model Predictive Control, Rehabilitation Engineering.

Università Campus Bio-Medico di Roma

Rome, Italy

BSC. IN INDUSTRIAL ENGINEERING

Sep 2021 - July 2024

- GPA: 30.17/30, graduated with full honors
- Relevant coursework: Applied Mechanics, Rational Mechanics Laboratory, Mechanics of solids and structures, Electronics, Bioengineering laboratory, Control of Digital Systems, Operating Systems and Computer Networks

Experience

Embotech AG

Zürich, Switzerland

SOFTWARE DEVELOPER AND MOTION PLANNING INTERN: MPC TEAM

Ma. 2026 - Aug. 2026

- Intern in the MPC team, working on **motion planning** for autonomous driving logistic vehicles. Responsible for new features, as part of the R&D of the company.

EPFL - Ecole Polytechnique Fédérale de Lausanne

Lausanne, Switzerland

RESEARCH INTERN

July - Sep 2024

- As a participant in the highly selective E3 program at EPFL, with a 2% acceptance rate, I was engaged in advanced research under the supervision of Dr. Josie Hughes. My work focused on **soft robotics**, particularly in **soft sensing** (proprioception and vision based control), and on a bimanual teleoperated robot for agricultural harvesting. One of our work has been accepted to IROS 2025.

Academic Projects (selected)

High-Fidelity Simulation for Soft Object Manipulation

Soft Robotics Lab- ETH

- Developing a learning pipeline for deformable-object manipulation that uses high-fidelity FEM simulation as a privileged expert. The approach combines trajectory optimization and MPC with policy distillation, and will be evaluated through sim-to-real deployment on a Franka robot.

Oct. 2026 - Apr. 2027

Nonlinear MPC for Flexible drones

Robotic Perception Group (RPG - UZH)

- Developed a **high-fidelity dynamics model** for a **flexible-joint quadrotor** and integrated it into a **multi-phase Model Predictive Contouring Control** framework. The controller combines flexible-joint, rigid-body, and point-mass models along the prediction horizon to enable long-horizon planning at **real-time rates**, and was validated in simulation and real flight.

Sep. 2025 - Jan. 2026

Multi Camera Gaussian Splat SLAM

Computer Vision and Geometry Group

- Me and my team synthesized a multi-camera SLAM system that uses **3D Gaussian Splatting** for robust localization, dense reconstruction, and novel view synthesis. By integrating **multiple fisheye cameras**, the system is designed to operate in challenging real-world environments and produce photorealistic maps. **Custom CUDA kernels** have been produced for the underlying NeRF NN.

Feb. 2025 - Jun. 2025

Academic Publications

- **Petitti, A.**, Rieter, R., Bauersfeld, L., Geles, I., and **Davide Scaramuzza** *Multi-Phase Model Predictive Quadrotor Contouring Control*, **Pre-print**.
- **Petitti, A.**, Obayashi, N., Bianchi, G. & Hughes, J. *Camera-tracked Soft Underwater Robot Enabling Robust Orientation Control for Maneuverability*, **Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)**, Hangzhou, China, Oct. 2025.